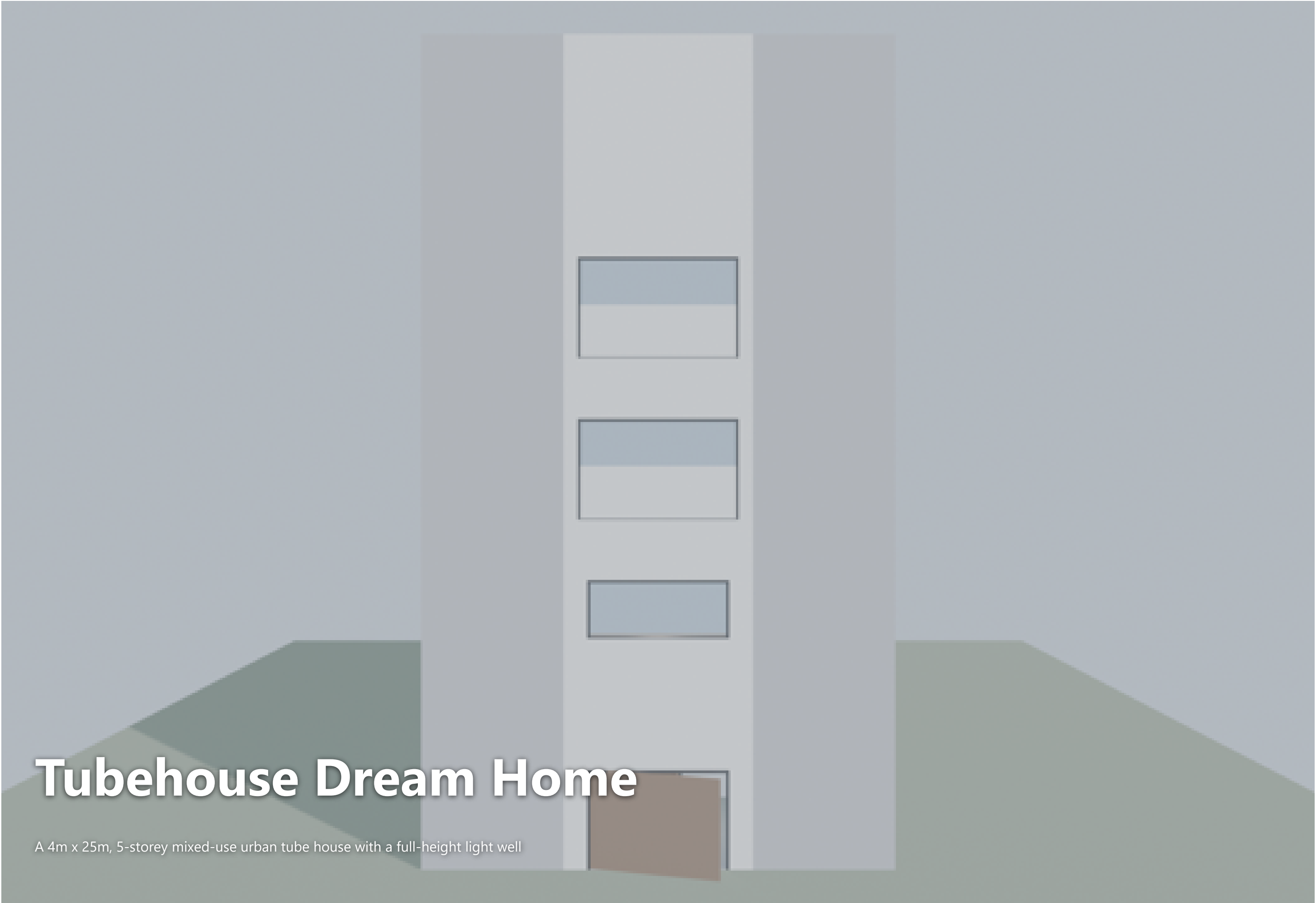




Tubehouse Dream Home

A 4m x 25m, 5-storey mixed-use urban tube house with a full-height light well

Design Intent

This brief sets out a 5-storey tube house on a narrow 4m x 25m urban plot, a typology built to bring daylight and cross-ventilation deep into a long, party-wall-bound footprint in a hot-humid climate.

A full-height light well runs through the middle of the tube beside the stair and elevator core, open to the sky from the ground floor to the roof terrace. It admits daylight to the deep mid-plan rooms on every level and drives stack-effect ventilation, drawing warm air up and out through the open roof void.

The ground and first floors are leased commercial space (garage-fronted retail/office bay and a lease studio), stacked below three private family floors: living/kitchen/dining, a master suite and kid's room, and a top-floor office and guest room opening onto a partial roof terrace.

Facade orientation is to be confirmed against the actual site; the glazed street elevation is designed for adjustable shading (louvres or deep overhangs) to be tuned once solar orientation is known.

The vertical circulation core (stair, elevator, light well) is shared and stacks identically on every storey, with a lockable family lobby at Floor 2 separating the private residence from the lease floors below.

Room Schedule

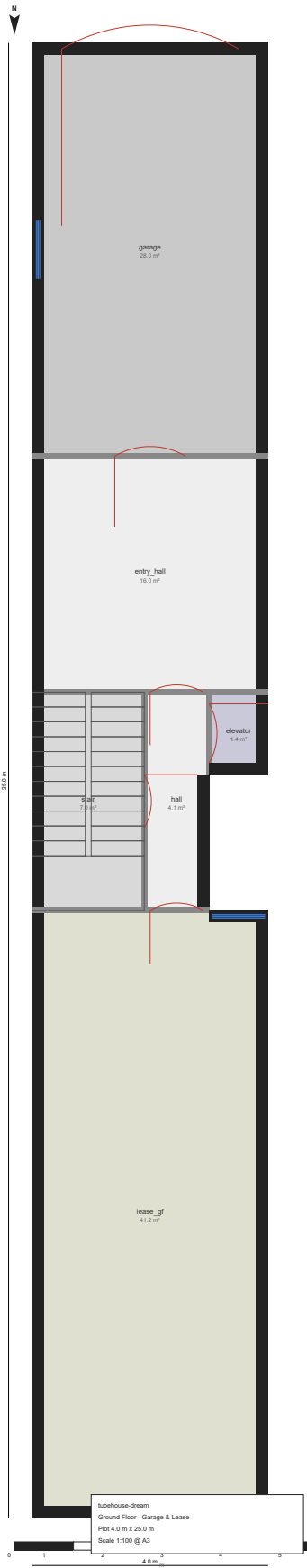
Floor	Room	Type	Area (m²)
Ground Floor - Garage & Lease	elevator	elevator	1.4
Ground Floor - Garage & Lease	stair	stairwell	7.0
Ground Floor - Garage & Lease	hall	hall	4.1
Ground Floor - Garage & Lease	garage	garage	28.0
Ground Floor - Garage & Lease	entry_hall	hall	16.0
Ground Floor - Garage & Lease	lease_gf	office	41.2
Floor 1 - Lease Studio	elevator	elevator	1.4
Floor 1 - Lease Studio	stair	stairwell	7.0
Floor 1 - Lease Studio	hall	hall	4.1
Floor 1 - Lease Studio	lease_f1	office	44.0
Floor 1 - Lease Studio	lease_f1_rear	office	35.2
Floor 1 - Lease Studio	wc_f1	bathroom	3.0
Floor 1 - Lease Studio	storage_f1	storage	3.0
Floor 2 - Living, Kitchen & Dining	elevator	elevator	1.4
Floor 2 - Living, Kitchen & Dining	stair	stairwell	7.0
Floor 2 - Living, Kitchen & Dining	hall	hall	4.1
Floor 2 - Living, Kitchen & Dining	balcony_f2	balcony	6.0
Floor 2 - Living, Kitchen & Dining	living_f2	living	38.0
Floor 2 - Living, Kitchen & Dining	kitchen_f2	kitchen	18.0
Floor 2 - Living, Kitchen & Dining	dining_f2	dining	15.2
Floor 2 - Living, Kitchen & Dining	wc_f2	bathroom	4.0

Floor	Room	Type	Area (m²)
Floor 2 - Living, Kitchen & Dining	utility_f2	storage	4.0
Floor 3 - Master Suite & Kid's Room	elevator	elevator	1.4
Floor 3 - Master Suite & Kid's Room	stair	stairwell	7.0
Floor 3 - Master Suite & Kid's Room	hall	hall	4.1
Floor 3 - Master Suite & Kid's Room	balcony_f3	balcony	6.0
Floor 3 - Master Suite & Kid's Room	master_f3	bedroom	24.7
Floor 3 - Master Suite & Kid's Room	ensuite_f3	bathroom	6.6
Floor 3 - Master Suite & Kid's Room	walkin_f3	storage	6.6
Floor 3 - Master Suite & Kid's Room	kid_bed_f3	bedroom	19.0
Floor 3 - Master Suite & Kid's Room	bath_f3	bathroom	3.6
Floor 3 - Master Suite & Kid's Room	laundry_f3	storage	4.6
Floor 3 - Master Suite & Kid's Room	flex_lounge_f3	living	14.0
Floor 4 - Office, Guest Room & Roof Terrace	elevator	elevator	1.4
Floor 4 - Office, Guest Room & Roof Terrace	stair	stairwell	7.0
Floor 4 - Office, Guest Room & Roof Terrace	hall	hall	4.1
Floor 4 - Office, Guest Room & Roof Terrace	terrace_f4	balcony	44.0
Floor 4 - Office, Guest Room & Roof Terrace	office_f4	office	20.0
Floor 4 - Office, Guest Room & Roof Terrace	guestbed_f4	bedroom	15.2
Floor 4 - Office, Guest Room & Roof Terrace	guestbath_f4	bathroom	3.0
Floor 4 - Office, Guest Room & Roof Terrace	storage_f4	storage	3.0

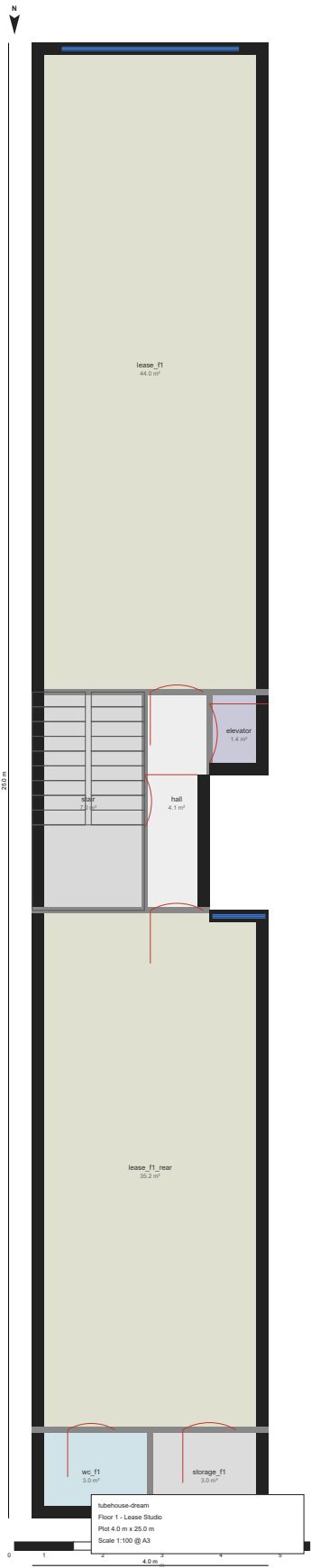
Floor Totals

Ground Floor - Garage & Lease	97.7 m²
Floor 1 - Lease Studio	97.7 m²
Floor 2 - Living, Kitchen & Dining	97.7 m²
Floor 3 - Master Suite & Kid's Room	97.6 m²
Floor 4 - Office, Guest Room & Roof Terrace	97.7 m²

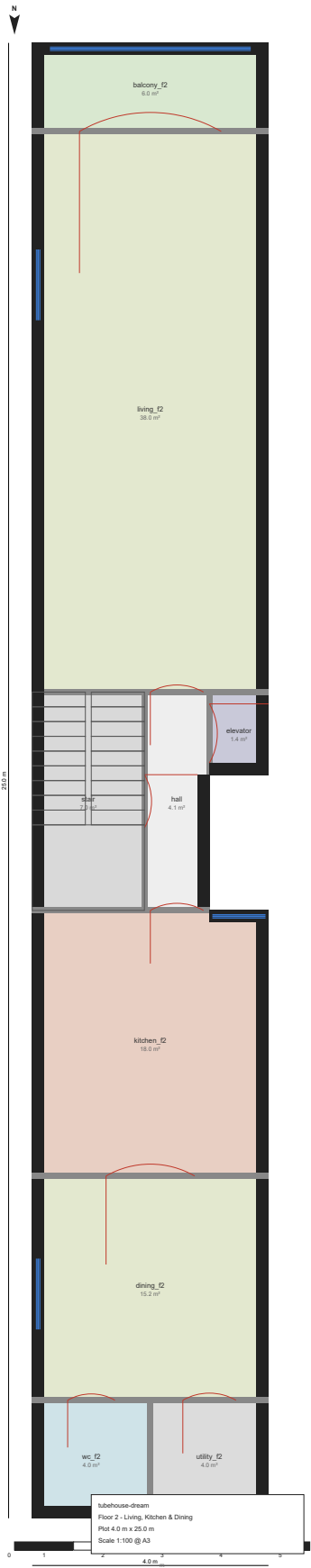
Ground Floor - Garage & Lease — Floor Plan



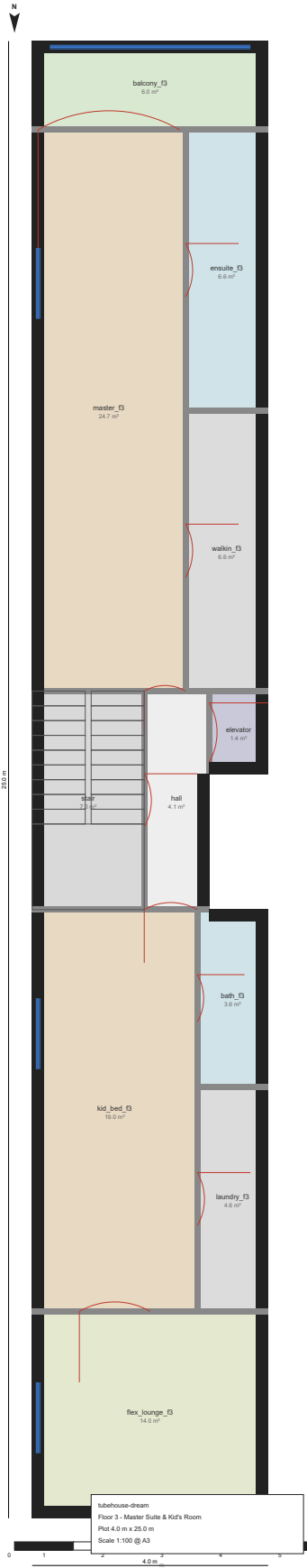
Floor 1 - Lease Studio — Floor Plan



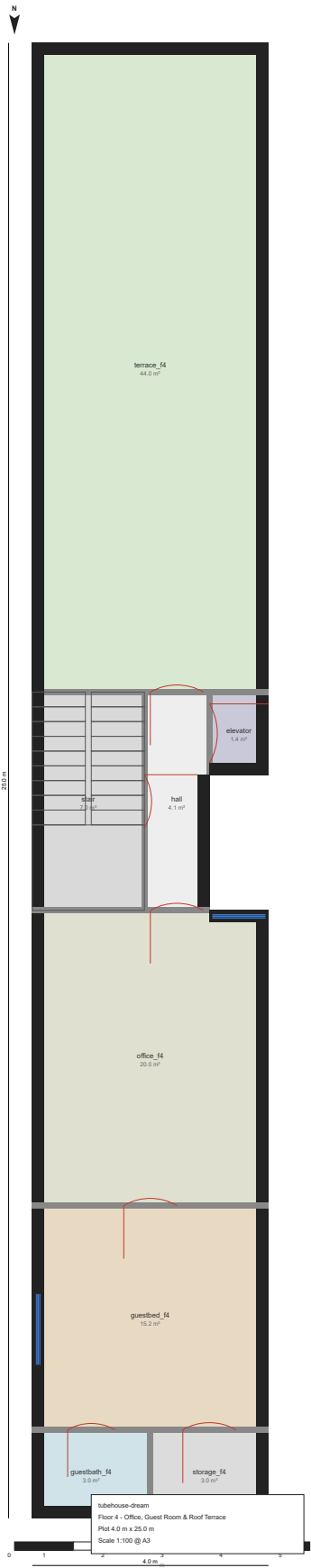
Floor 2 - Living, Kitchen & Dining — Floor Plan



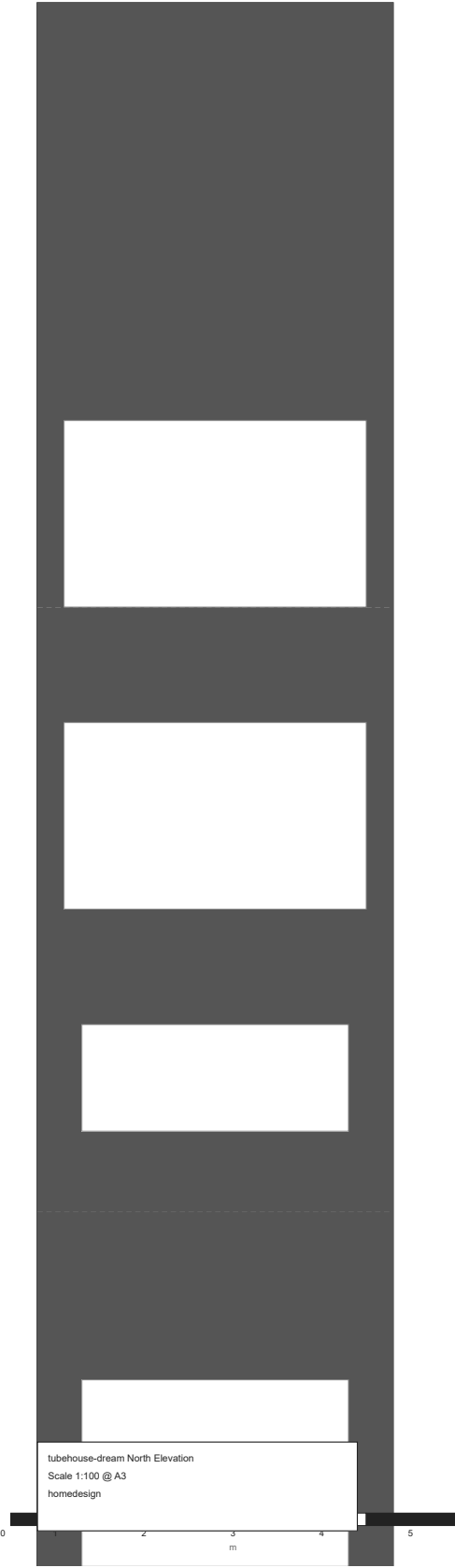
Floor 3 - Master Suite & Kid's Room — Floor Plan



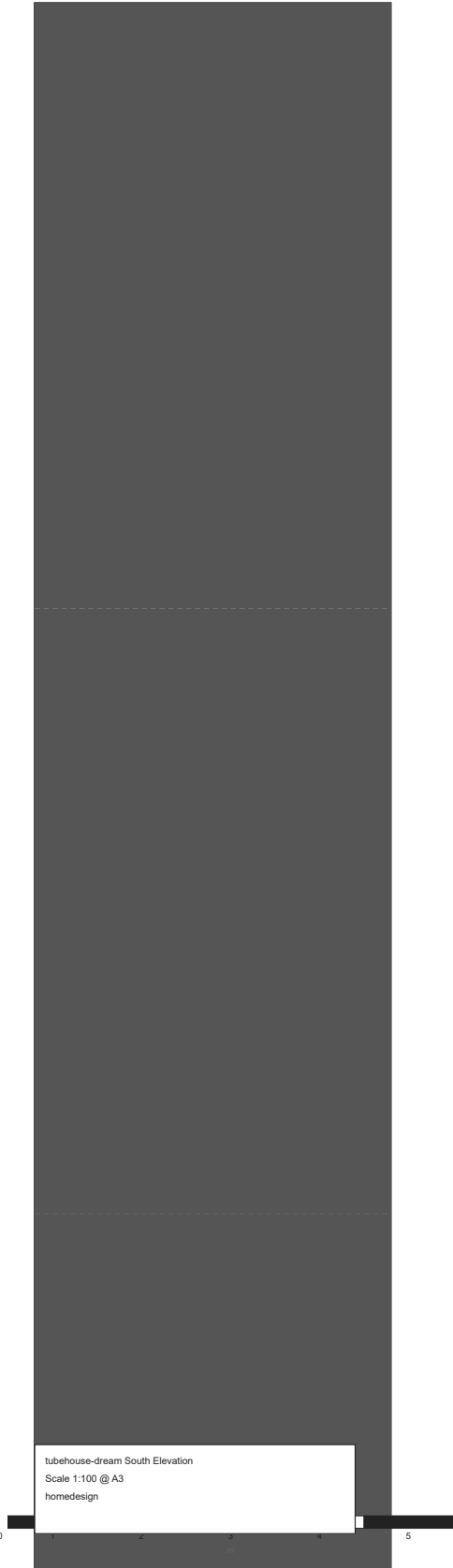
Floor 4 - Office, Guest Room & Roof Terrace — Floor Plan



North Elevation



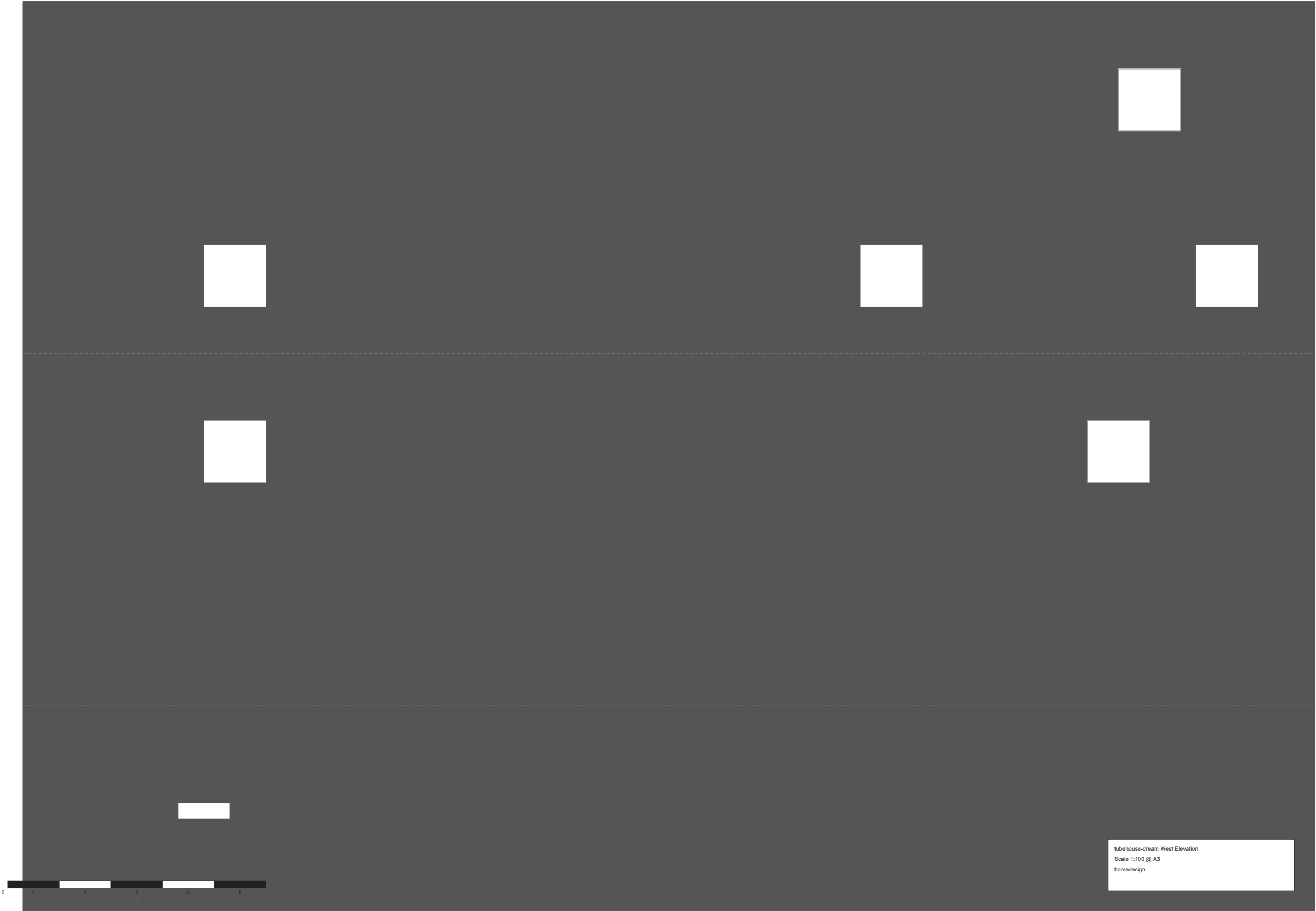
South Elevation



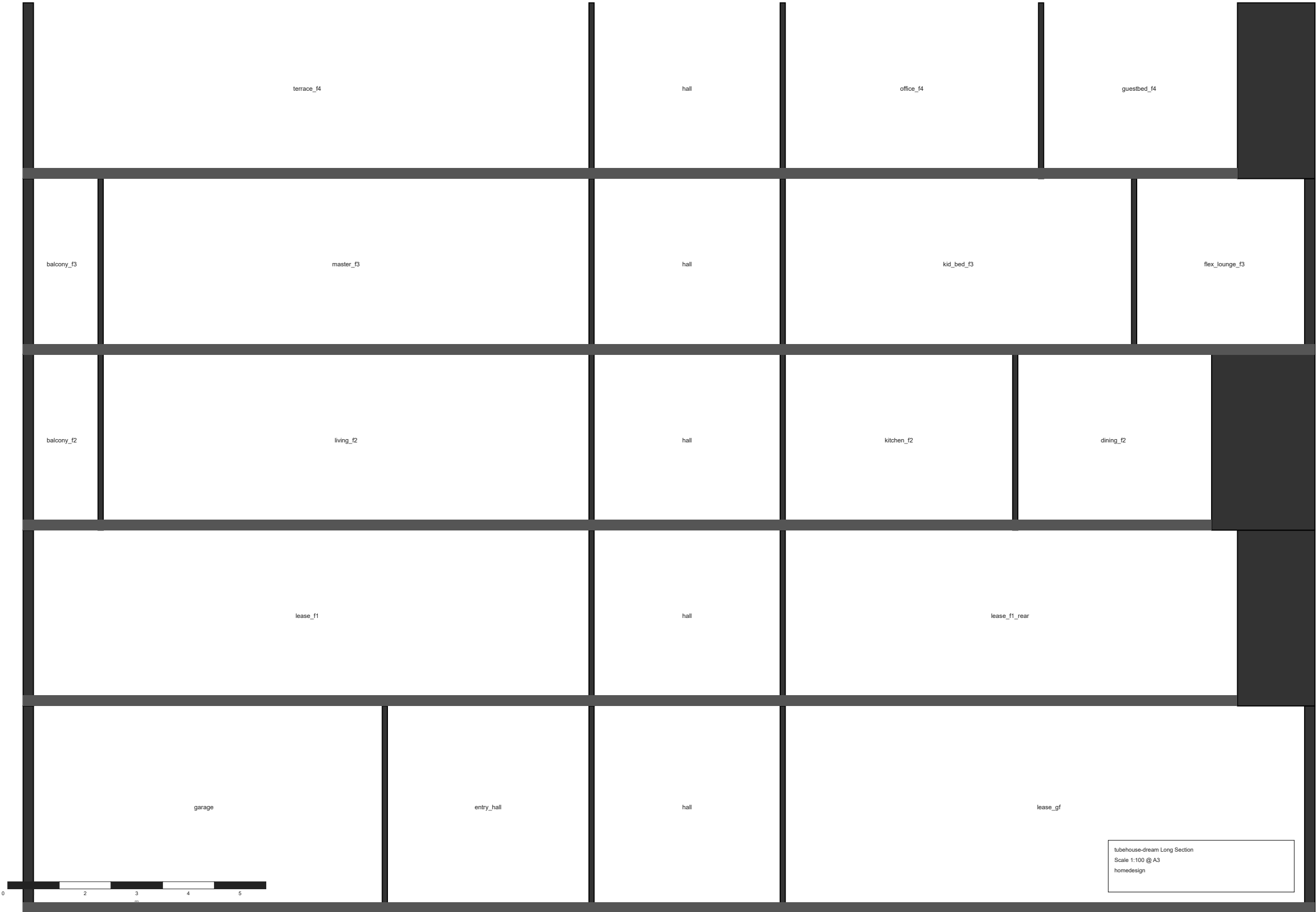
East Elevation



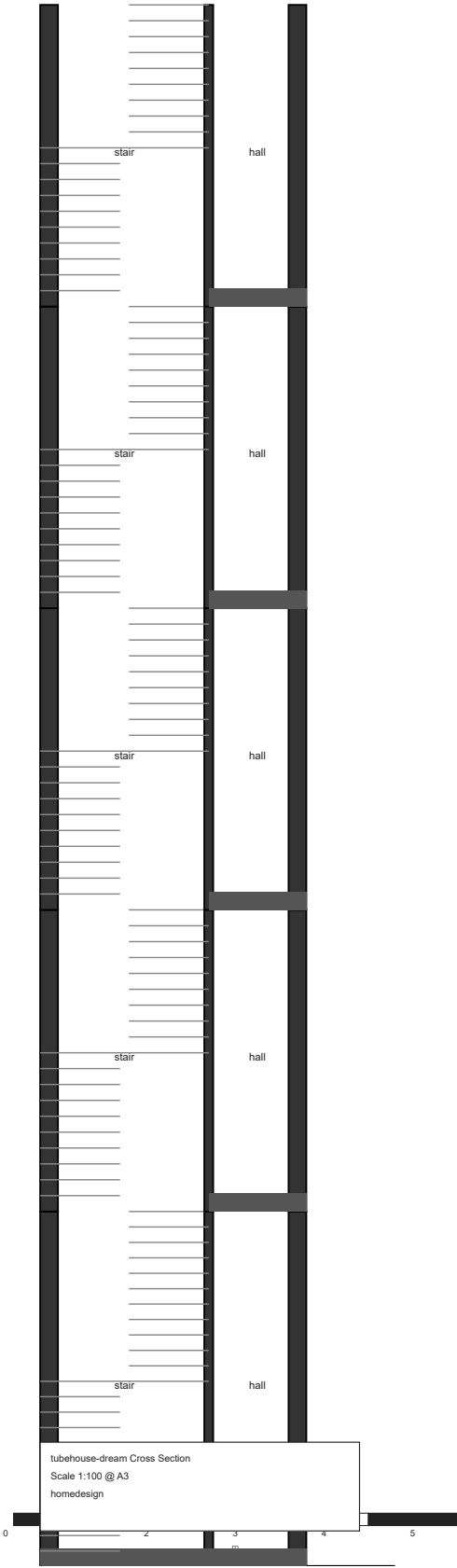
West Elevation



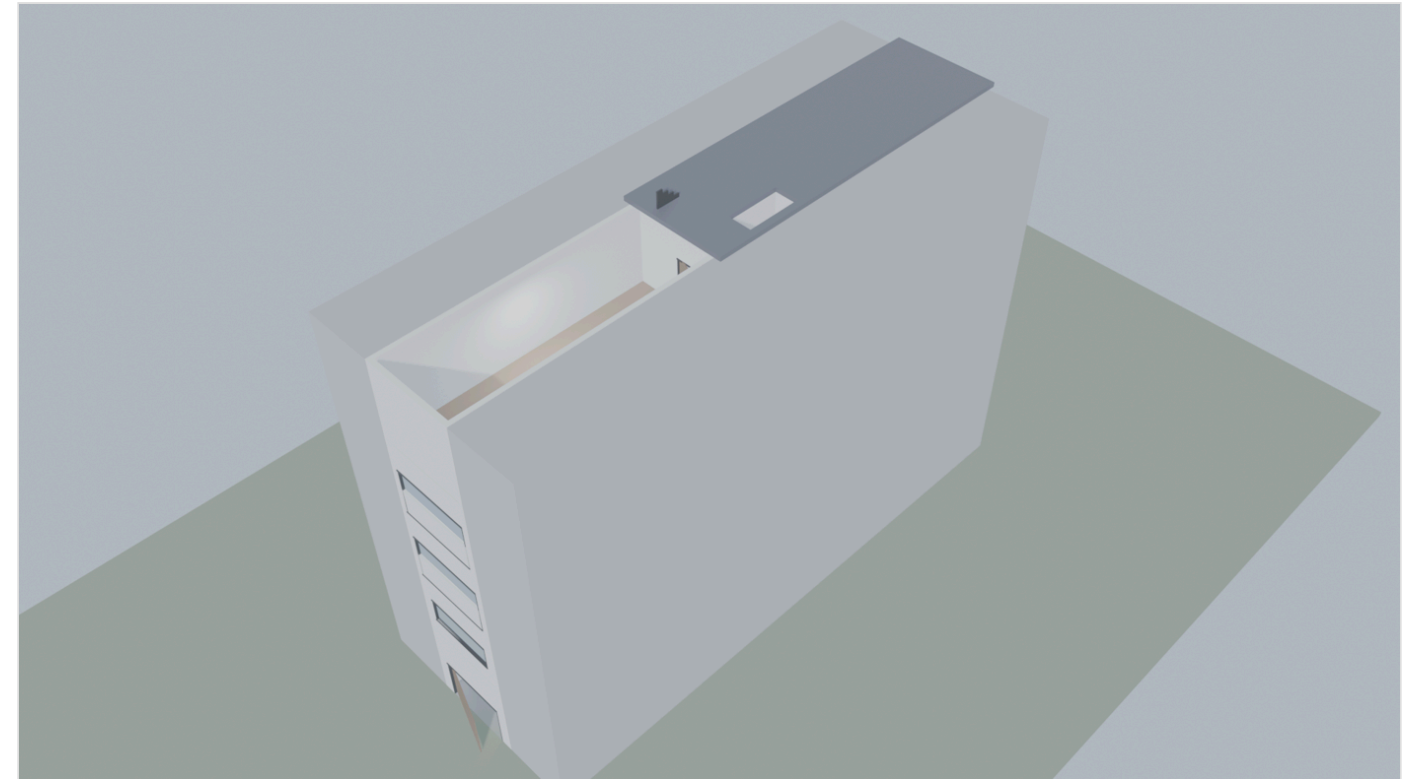
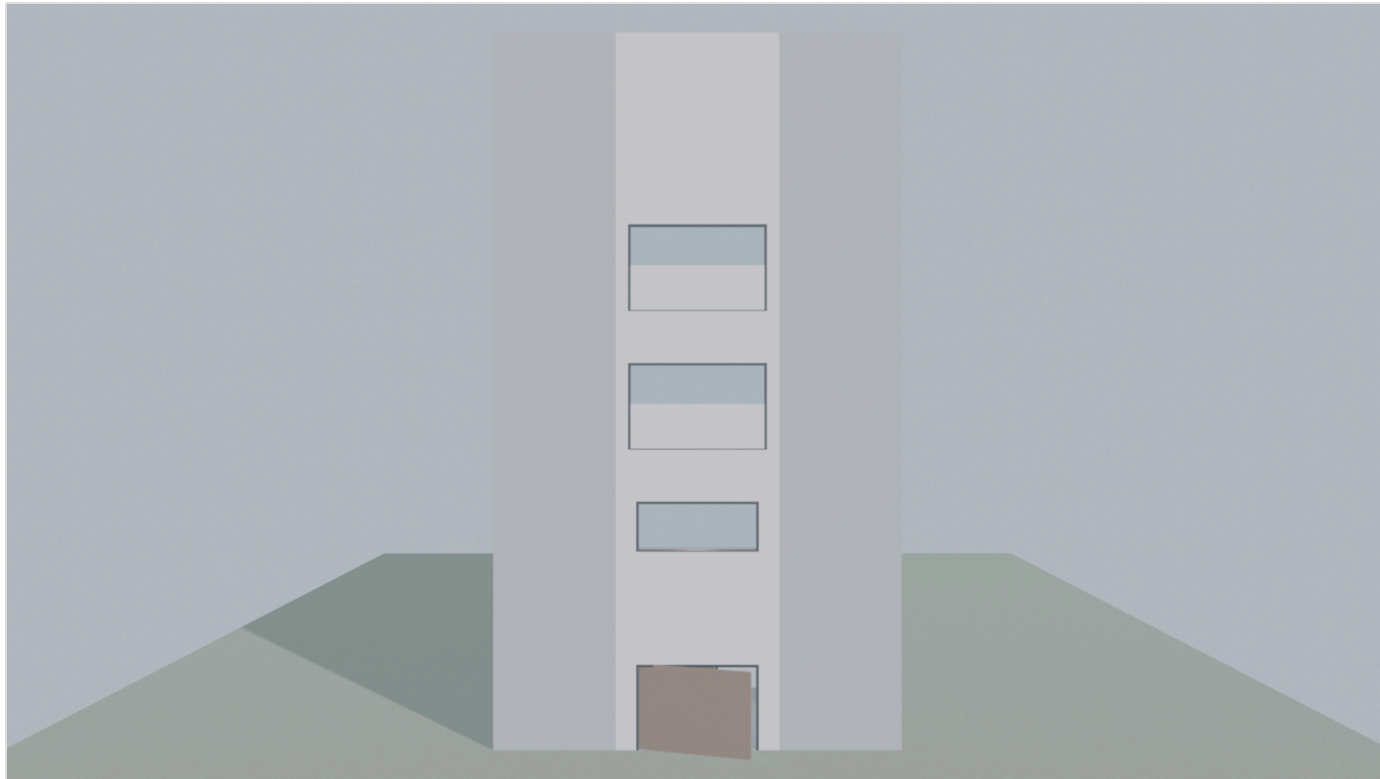
Long Section



Cross Section



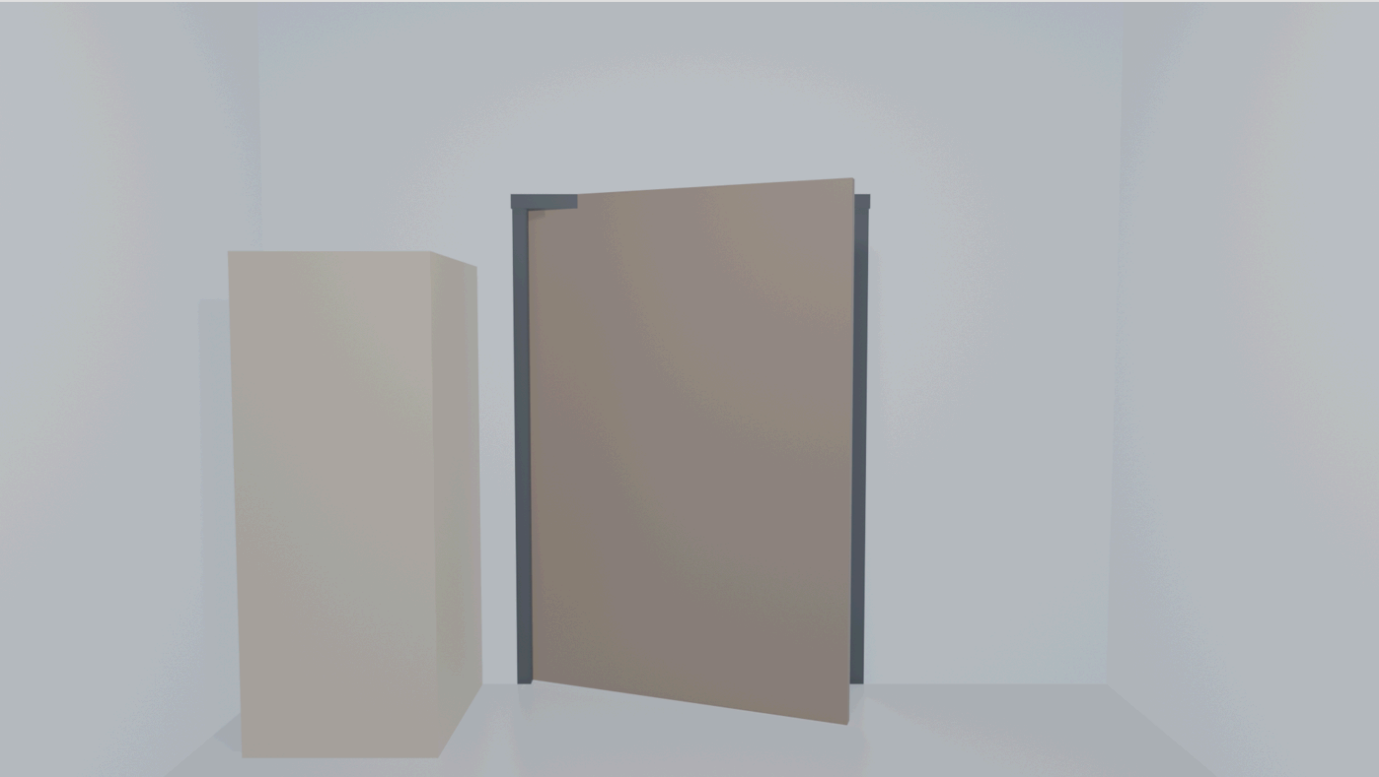
Renders



Renders



Renders



Renders



Renders



Requirements for the Architect

- Passenger elevator, family-sized (~1.2m x 1.5m shaft), stopping at all 5 levels.
- Light-well glazing (where rooms face the well) must be operable for ventilation and detailed for condensation control given the hot-humid climate.
- Light well requires a drainage plan at its ground-floor base (roof rainwater falling through the open void) and waterproofing at every floor edge that abuts it.
- Party walls (both long sides and rear) are structural masonry/concrete with no openings; all glazing is on the street facade or into the light well.
- A vertical services riser (electrical, plumbing, data) should run alongside the stair/elevator core, accessible from each floor's core zone.
- Facade orientation is not yet fixed to a real site; confirm solar orientation before finalizing shading design for the street-facing glazing.
- Lease floors (ground + Floor 1) need access separation from the family residence above: a lockable lobby door at Floor 2 restricts the shared stair/elevator core to residents beyond that point.
- No cost or budget content is included in this brief; it is a design-intent and requirements document only.

Door & Window Schedule

ID	Floor	Type	Width	Sill	Head	Between
F0_O000	0	door	1000	0	2100	exterior / ?
F0_O001	0	door	900	0	2100	exterior / ?
F0_O002	0	door	3000	0	2100	exterior / ?
F0_O003	0	door	1200	0	2100	exterior / ?
F0_O004	0	door	900	0	2100	exterior / ?
F0_O005	0	door	900	0	2100	exterior / ?
F0_O006	0	window	900	1000	2100	exterior / ?
F0_O007	0	window	1000	1800	2100	exterior / ?
F1_O000	1	door	1000	0	2100	exterior / ?
F1_O001	1	door	900	0	2100	exterior / ?
F1_O002	1	door	900	0	2100	exterior / ?
F1_O003	1	door	900	0	2100	exterior / ?
F1_O004	1	door	800	0	2100	exterior / ?
F1_O005	1	door	900	0	2100	exterior / ?
F1_O006	1	window	900	900	2100	exterior / ?
F1_O007	1	window	3000	900	2100	exterior / ?
F2_O000	2	door	1000	0	2100	exterior / ?
F2_O001	2	door	900	0	2100	exterior / ?
F2_O002	2	door	2400	0	2100	exterior / ?
F2_O003	2	door	900	0	2100	exterior / ?
F2_O004	2	door	900	0	2100	exterior / ?
F2_O005	2	door	1500	0	2100	exterior / ?
F2_O006	2	door	800	0	2100	exterior / ?
F2_O007	2	door	900	0	2100	exterior / ?
F2_O008	2	window	3400	0	2100	exterior / ?
F2_O009	2	window	900	900	2100	exterior / ?

ID	Floor	Type	Width	Sill	Head	Between
F2_O010	2	window	1200	900	2100	exterior / ?
F2_O011	2	window	1200	900	2100	exterior / ?
F3_O000	3	door	1000	0	2100	exterior / ?
F3_O001	3	door	900	0	2100	exterior / ?
F3_O002	3	door	2400	0	2100	exterior / ?
F3_O003	3	door	700	0	2100	exterior / ?
F3_O004	3	door	900	0	2100	exterior / ?
F3_O005	3	door	900	0	2100	exterior / ?
F3_O006	3	door	900	0	2100	exterior / ?
F3_O007	3	door	800	0	2100	exterior / ?
F3_O008	3	door	900	0	2100	exterior / ?
F3_O009	3	door	1200	0	2100	exterior / ?
F3_O010	3	window	3400	0	2100	exterior / ?
F3_O011	3	window	1200	900	2100	exterior / ?
F3_O012	3	window	1200	900	2100	exterior / ?
F3_O013	3	window	1200	900	2100	exterior / ?
F4_O000	4	door	1000	0	2100	exterior / ?
F4_O001	4	door	900	0	2100	exterior / ?
F4_O002	4	door	900	0	2100	exterior / ?
F4_O003	4	door	900	0	2100	exterior / ?
F4_O004	4	door	900	0	2100	exterior / ?
F4_O005	4	door	800	0	2100	exterior / ?
F4_O006	4	door	900	0	2100	exterior / ?
F4_O007	4	window	900	900	2100	exterior / ?
F4_O008	4	window	1200	900	2100	exterior / ?

Quantity Take-Off

Storey	GFA (m²)	Ext. wall (m)	Partition wall (m)	Doors	Windows
Ground Floor - Garage & Lease	97.7	50.8	5.7	6	2
Floor 1 - Lease Studio	97.7	51.0	7.3	6	2
Floor 2 - Living, Kitchen & Dining	97.7	51.0	8.0	8	4
Floor 3 - Master Suite & Kid's Room	97.7	50.8	22.7	10	4
Floor 4 - Office, Guest Room & Roof Terrace	97.7	51.0	7.4	7	2
Built envelope (W x D m)	4.0	25.0	0.0		
Total	488.5	254.6	51.1	37	14

Handover Files

The following machine-readable files accompany this brief:

- tubehouse-dream.json (source spec, JSON)
- tubehouse-dream_f0.dxf (Ground Floor - Garage & Lease, CAD/DXF)
- tubehouse-dream_f1.dxf (Floor 1 - Lease Studio, CAD/DXF)
- tubehouse-dream_f2.dxf (Floor 2 - Living, Kitchen & Dining, CAD/DXF)
- tubehouse-dream_f3.dxf (Floor 3 - Master Suite & Kid's Room, CAD/DXF)
- tubehouse-dream_f4.dxf (Floor 4 - Office, Guest Room & Roof Terrace, CAD/DXF)